Paleomagnetic Results From the Upper Triassic of East Greenland

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Study of a 12.2-m inclined core from within the Malmros klint member of the Fleming Fjord formation at 71.73°N, 23.42°W reveals a N-R-N-R-N polarity sequence within the core, where N is normal and R is reversed. The lower two normal zones are each less than 1 m thick and are separated by the 1-m-thick lower reversed zone. The upper transition (R to N) is abrupt. A paleomagnetic pole position at 103°E, 34°N has been calculated from selected reversed samples. This pole is considered well determined in that (1) it is almost exactly 180° opposite a pole calculated from normal polarity samples and (2) samples involved in the determination appear to be well demagnetized when they are plotted on orthogonal vector demagnetization plots. It does not agree, however, with the position calculated by rotating the mean North American Upper Triassic paleomagnetic pole about the Greenland–North American rotation axis by an amount sufficient to open the Davis Strait in post-Triassic time. The calculated pole position is approximately 20° ESE of a similar pole predicted from European data, although severe selection of the data would tend to halve the distance to the predicted European pole. The paleomagnetic pole derived from the Fleming Fjord formation in this study supports the suggestion that Greenland was attached to the European plate rather than to the North American plate in the Late Triassic and was located considerably south of its present location at that time.

Late in the field season of 1972 a 12.2-m continuously drilled inclined core was collected by D. Leythaeuser and K. W. Bay from a site 200 m north of the first side valley on Allday Dal, a tributary to Oersted Dal in Jameson Land, east Greenland. A general location map is shown in Figure 1. The site coordinates are 71°44'N, 23°25'W. The samples were collected from the lower middle portion of the Malmros klint member of the Fleming Fjord (formerly Cape Biot) formation, the top of the core being located very roughly 75 m above the base of the member. It was originally planned to drill down to the base of the Malmros klint member and thus acquire a reversal sequence covering a substantial portion of, if not all of, Carnian time, but the threat of early snow forced a halt to drilling.

This paper reports the results of 1200 measurements on 208 specimens cut from 134 individually oriented samples from that core.

GEOLOGY

The Malmros klint member of the Fleming Fjord formation (formerly the Fleming Fjord member of the Cape Biot formation [*Perch-Nielsen et al.*, 1972]) is the middle member of the formation. It is overlain by the Oersted Dal member, a series of mudstones, sandstones, and arkosic conglomerates topped by limestone, dolomite, and a bone bed of Rhetic age. It is underlain by the Edderfugledal member, a siltstone with stromatolitic and oolitic layers [*Perch-Nielsen et al.*, 1972]. The Malmros klint member itself is a series of dark red to red brown (5YR-5/2, 5YR-3/2) and some purplish (5RP-4/2) mudstones. Publications previous to the work of Perch-Nielsen et al. [e.g., *Grasmück and Trümpy*, 1969: *Trümpy*, 1961] placed the Kap Seaforth member of the Gipsdalen formation, a series of predominately gypsum and dolomite, in the Fleming Fjord and included the Edderfugledal within it. For more detailed descriptions of the lithologies, we would refer readers to *Grasmück and Trümpy* [1969]. Fossil phyllopods from the base of the Malmros klint are of lower-most Upper Triassic age, and ostracods within the lower portion of the Oersted Dal member are suggestive of a Rhetic age. These ages, along with the definite Rhetic bone bed at the top of the Oersted Dal, bracket the Malmros klint in the Carnian-Norian age range. Paleoenvironmental interpretation of the members of the Fleming Fjord relates the Edderfugledal member (and Kap Seaforth) to a shallow evaporite basin and the Malmros klint member to associated tidal mud flat environments, and the Oersted Dal records a period of Fluvial deposition [*Surlyk et al.*, 1971] and return to marine evaporitic conditions.

The Malmros klint member is approximately 200 m thick in the area of Cape Biot and thins westward, being somewhat thinner in the study area.

The intrusion of dolerite sills and dikes during the Tertiary is an important factor in the discussion of the red bed magnetics. The sills are most typically emplaced between Oersted Dal and Kap Stewart. Although these intrusions show little evidence of contact metamorphism, their emplacement almost certainly raised regional temperatures enough to add small amounts of fairly stable low-temperature secondary thermal or viscous remanent magnetizations to the original remanence. Comparison of *Tarling*'s [1966] studies and pole positions for the Tertiary igneous rocks with our results from this study leads us to believe that the pole published by *Bidgood and Harland* [1961] for the Cape Biot (Fleming Fjord) formation actually records a Tertiary, not a Triassic, field direction. A large sill outcrops in the valley wall 1.25 km to the east of our sampling site and appears to have overlain the sample locality [*Bromley et al.*, 1970, map 2]. The demagnetization behavior observed in our samples is, however, only a hint, not proof, that the samples have acquired a secondary thermal remanent magnetization. Furthermore, *Tarling* [1966] and *Tarling and Otulana* [1972] have suggested that most of the Tertiary igneous rocks were implaced during a short period of reversed polarity, whereas the low-temperature moments observed in our rocks are mixed but predominantly normal, reversed overprints being present in some samples.

TECHNIQUES

Samples were collected by means of a portable core (Winkie) drill produced by J. K. Smit and Sons, Toronto,

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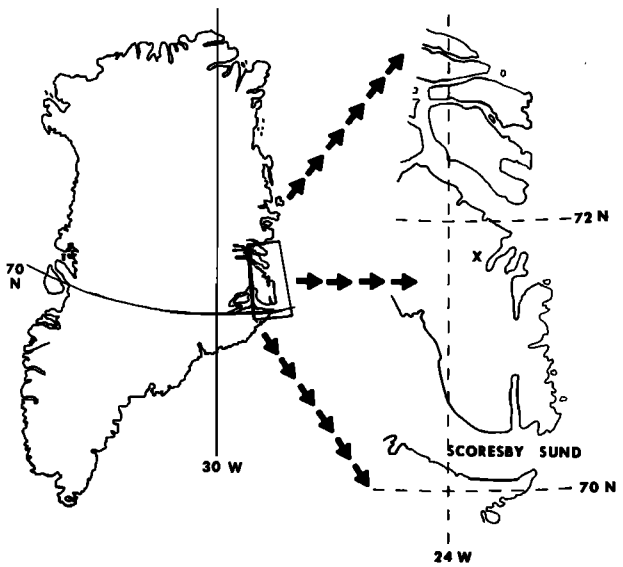


Fig. 1. General location map. Sampling site is denoted by the cross shown in the enlarged area.

Canada, and a magnesium drill pipe. The orientation of the drill hole was 72° dip in a N65°E dip direction. The dip direction was obtained by a magnetic compass. This dip direction has been corrected to N61.5°E in our programs to compensate for a systematic orientation error caused by the difference in the hole dip direction and the dip direction of the rocks at the drilling site. Structural correction for a dip of 7° in the direction N79°E has been made on all data. Upon their return to the laboratory all samples were oriented by using the elliptical bedding plane section in a method described by *Baag and Helsley* [1974]. Samples were then cut into 2.5-cm-long sections and measured on a PAR SM-1 spinner magnetometer. Demagnetization was accomplished by heating the samples in a noninductively wound furnace and cooling them in a mu-metal cylinder inside a magnetically shielded room. Maximum field exposure for any cooling sample was less than 5 γ. All samples were heated to 275°, 425°, 550°, 608°, and either 627° or 647°C to allow for individual interpretation of the polarity of each sample on the basis of orthogonal vector demagnetization diagrams. In addition, over half the samples were first heated and measured at a 150°C demagnetization step.

Demagnetization and Poles

Average declination and inclination for the complete set of natural remanent magnetizations (NRM) was N35.6°E and 68.7° down after structural correction. The average NRM declination of samples shown to be normally magnetized is 38.5°, and the inclination 65.4°. Reverse samples are heavily overprinted with a normal component, and their NRM's

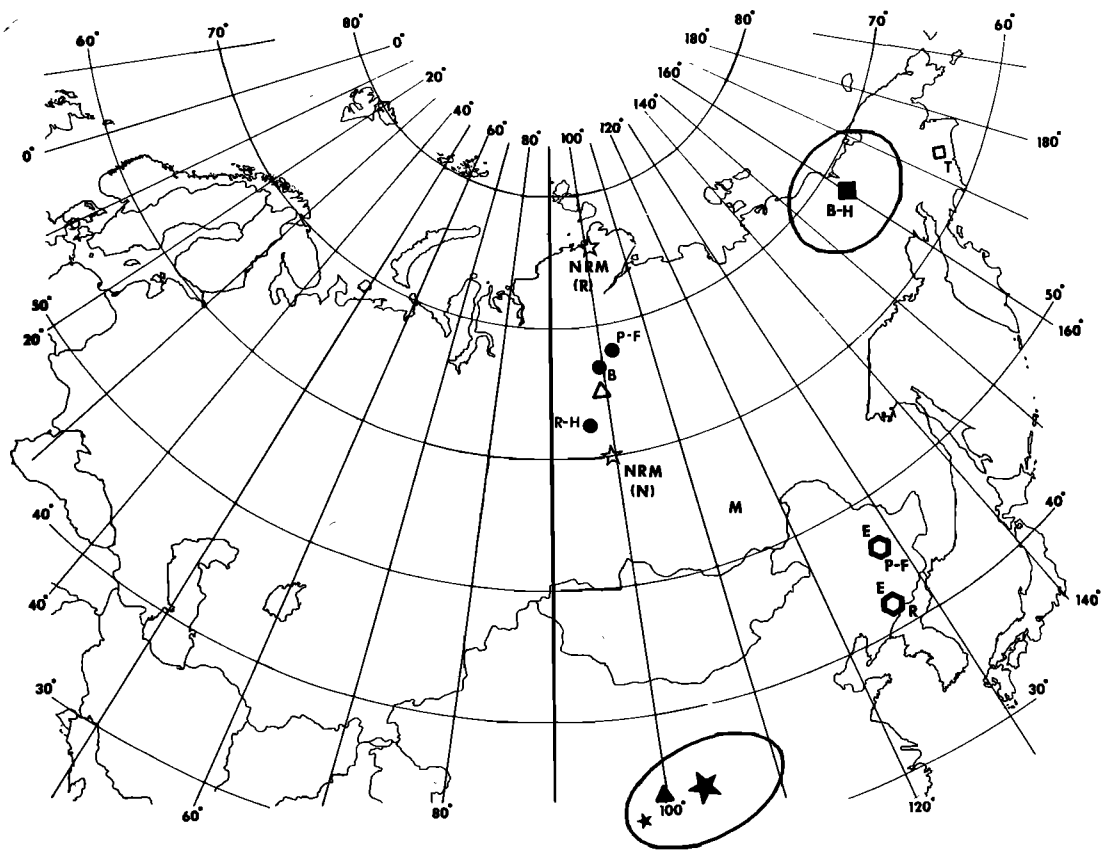


Fig. 2. Comparison of Upper Triassic pole positions from North America, Europe, and Greenland. Solid circles are rotated North American Upper Triassic mean poles [*Phillips and Forsyth*, 1972; *Beck*, 1972; *Reeve and Helsley*, 1972]. Hexagons are rotated European mean poles [*Phillips and Forsyth*, 1972; *Roy*, 1972]. The solid square is the Fleming Fjord pole [*Bidgood and Harland*, 1961]. The open square is *Tarling*'s [1966] Tertiary igneous pole from east Greenland. Open starts are the NRM's of normal (N) and reversed (R) intervals described in this study. The large solid star denotes the primary pole position and the confidence limits derived for the Fleming Fjord formation, selected reversed data from this study being used. The small solid star is the pole position based on unselected normal samples from this study. The open triangle is the average of all NRM's from Fleming Fjord. The solid triangle is the 627°–647°C demagnetization step for all data. The rotated Lower Triassic pole for the Moenkopi formation is denoted by M.

average a 24.8° declination and a 74.6° inclination. For comparison with the final high-temperature data, pole positions based on the NRM of the samples are plotted in Figure 2. Stepwise thermal demagnetization caused little change in direction until the samples had been heated to 425°C for periods of approximately 15 min. In pilot studies on a group of 30–40 samples of both normal and reversed polarity, heatings at 150°, 275°, 350°, and 425°C produced no major changes in direction except for an initial movement of 5° to the southeast. Poles calculated for the three low temperatures form a small triangle 3°–4° on a side approximately 104°E, 59°N. Upon demagnetization of the entire population at 400°C it becomes clear that at least some samples are reversed, and the good (if somewhat elongate) cluster (Figure 3) at 275°C becomes a definite streak. Figure 4 shows stereographic plots of declination and inclination for two of the higher-temperature demagnetizations. Grouping and separating reversed and normal polarity samples are still incomplete at the 550°C step and to some extent even in the 627°–647° plot. In this interval the number of samples scattered about the edges of the group is still large, but normal and reversed polarity clusters are well separated and defined, and a number of samples have ceased to show direction changes in the interval between the heating at 608°C (not plotted) and the high temperatures.

To calculate a pole position from the data, a variation of the method described by *Helsley* [1973] has been used to pick the most stable samples. This technique consists of picking the samples from streaked data in groups on the basis of their position in the original NRM distribution. Because no initial streak was present, this selection has instead been made on the 550°C demagnetized sample data. In theory, Helsley's method rests on the assumption that those samples closest to the expected pole position at (or near) the end of the NRM streak will be the most stable and more likely to give true field directions in comparison with those at the opposite end of the streak, which have probably acquired at least some stable components in the secondary direction.

In this paper we have followed a similar if somewhat less esthetic course in assuming that those samples nearest the expected direction at the 550°C demagnetization step have the strongest primary magnetizations and are therefore the most likely to yield true directions at the higher demagnetization steps. We have therefore arbitrarily chosen to use only the high-temperature demagnetizations of those samples that had reached the negative third quadrant by 550°C in order to calculate a pole position.

The pole so calculated is located at 103.2°E, 34°N with an α_{95} of 5.4° (*dp* 3.8°, *dm* 6.4), a Fisher's estimate of precision k of 24.8, and a sample size of 30. These values yield a paleolatitude of 22°N, which is compared with the paleolatitudes given by *Roy* [1972] for North America and Europe in Figure 5.

This pole is shown in Figure 2 along with a set of hypothetical 'expected' positions for comparison. These expected positions have been obtained by rotating the mean North American Triassic poles of *Beck* [1972], *Phillips and Forsyth* [1972], and *Reeve and Helsley* [1972] about the North American–Greenland rotation axis of *Bullard et al.* [1965] at 94.4°W, 70.5°N by an amount (18.0°) sufficient to close the Davis Strait. Likewise, the average Triassic poles of *Roy* [1972] and *Phillips and Forsyth* [1972] have been rotated by 22° about a pole at 96.5°E, 73°N to bring Europe against Greenland. The agreement of the pole from the Fleming Fjord formation with the rotated North American pole is

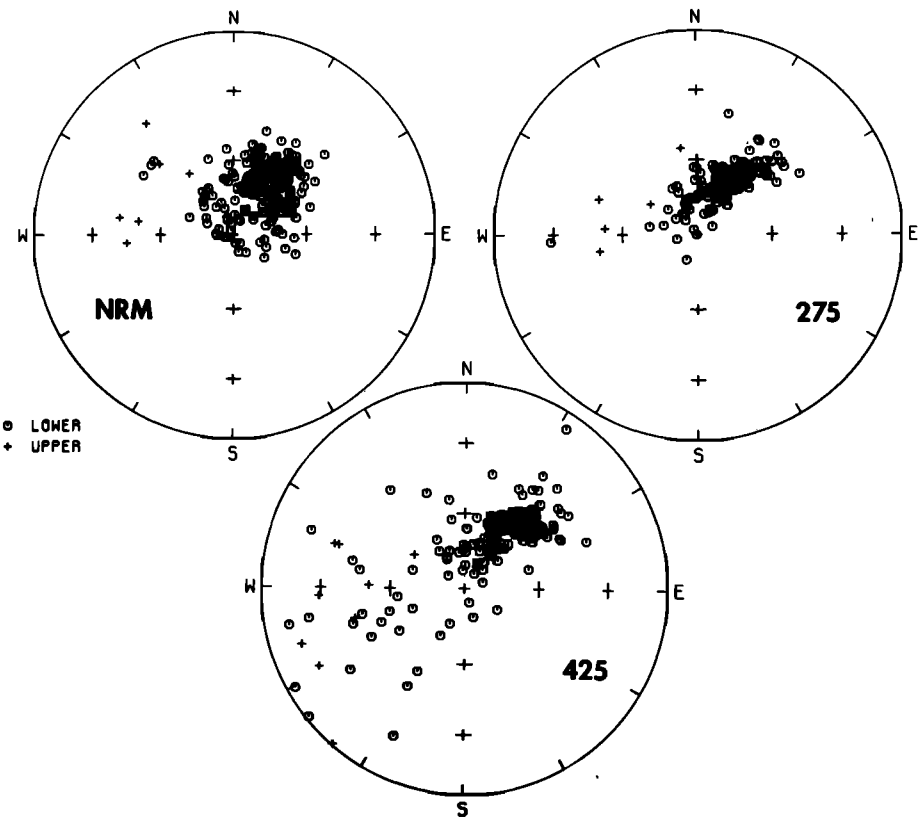


Fig. 3. Structurally corrected stereographic plots of declination and inclination for all samples whose NRM was measured and for all samples demagnetized at 275° and 425°C. The solid square in each plot represents the present-day magnetic field direction at the sampling site.

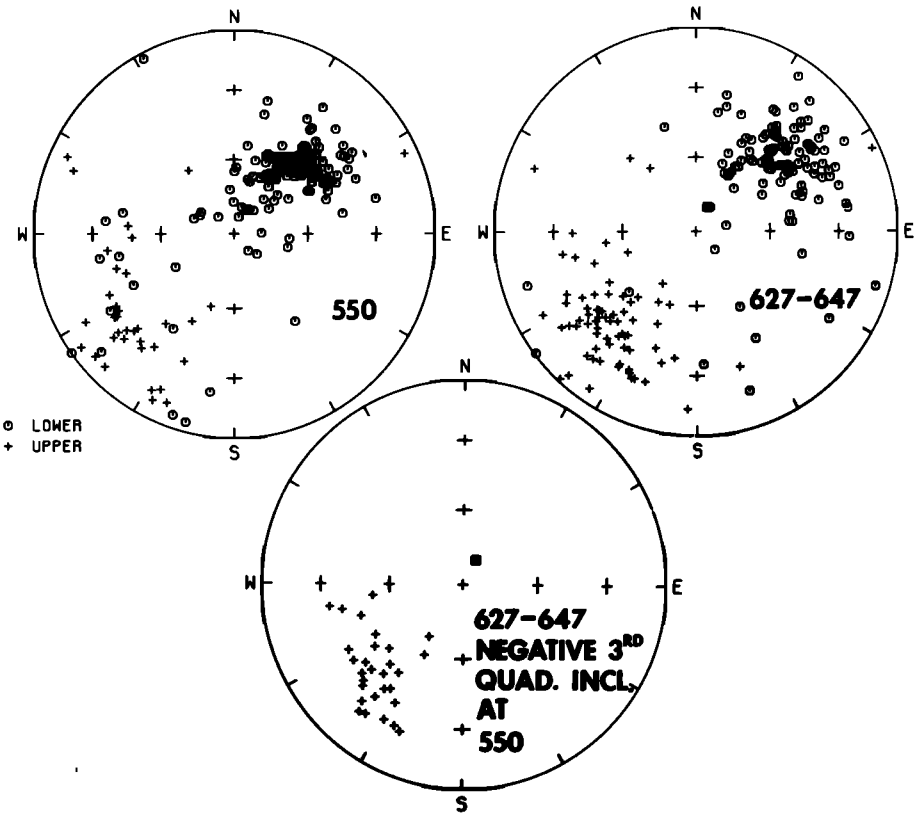


Fig. 4. Structurally corrected stereographic plots of declination and inclination of all samples demagnetized at 550°C and either 627° or 647°C. In the bottom plot all samples except those that had reached the upper third quadrant by 550°C have been removed. The solid square in each plot represents the present-day magnetic field direction at the sampling site.

poor, and agreement is only somewhat better with the European poles.

The failure of the pole for the Fleming Fjord to agree with its predicted position and, indeed, the failure of the North American and European poles to coincide are puzzling. Some of the difference may be due to age discrepancies, since the means of *Beck* [1972] and *Reeve and Helsley* [1972] included only Upper Triassic rocks (some of which are now believed to be Lower Jurassic in age [*Cornet et al.*, 1973]), whereas the European data are based on both Lower and Upper Triassic rocks. Rotating the mean for the Lower Triassic Moenkopi formation at 104.6°E, 54.4°N [*Helsley and Steiner*, 1974] gives an estimate of the minimum separation between the Lower Triassic North American mean and the European means due to the age differences alone. The remaining distance should probably be considered as a reflection of the uncertainty and quality of the European data. We consider the North American pole well determined, although heavily weighted by the Upper Triassic–eastern United States igneous rocks. The Fleming Fjord also lies considerably to the south of the North American pole, however, and is based on well-demagnetized samples.

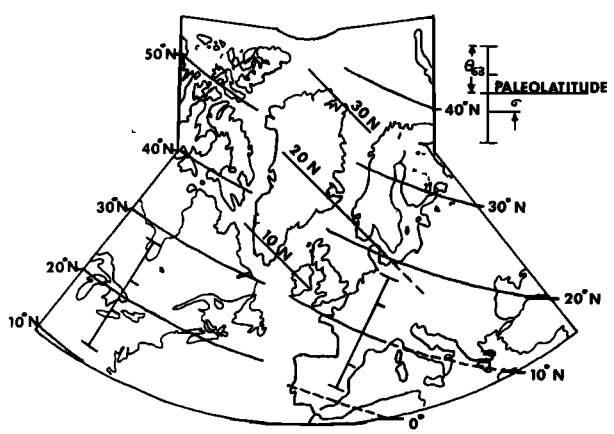


Fig. 5. Paleolatitude map of continents around the North Atlantic, modified from *Roy* [1972].

To explain the discrepancies between the European and North American poles, *Roy* [1972] proposed a right lateral strike slip fault between North America and Greenland that would have been located in a proto-Atlantic and been active prior to the Upper Triassic. Although we regard the existence of this fault and the 20° northward displacement proposed by Roy as being severely limited by the correlation of the geology on each side of the Davis Strait and by the proposed Middle Jurassic or younger initial rifting [*Hallam*, 1971; *Clarke and Upton*, 1971; *Watt*, 1969; *Laughton*, 1972], our results are consistent with Roy's theory. Furthermore, if the European pole positions are accurate, the rough agreement in paleolatitude between them and the Greenland pole supports the hypothesis that Greenland was attached to the European block.

A second possible explanation for the discrepancies between the North American and Greenland poles is that the field recorded by our samples was anomalous in direction. This hypothesis appears unlikely, however, in that it would require the field to remain anomalous during at least three different polarity periods.

REVERSALS

As was previously pointed out, high-temperature demagnetization of the core shows the presence of both nor-

mal and reversed polarity. A stratigraphic plot of declination, inclination, and intensity of the samples demagnetized at high temperature is shown in Figure 6, and a similar plot of the NRM is shown in Figure 7. The plots are corrected for the inclination of the core, ground level being the 12-m mark. From these figures, as well as Figures 3 and 4, the heavy overprinting of the reversed samples is clearly evident, only the slightest hint of a possible reversal being revealed between 4.5 and 7 m in the NRM plot.

The upper 4 m of the core exhibits normal polarity. The four reversed specimens in this zone are each small single samples with large orientation errors. The first reversal below ground level occurs just below the 8-m mark. The transition appears to be abrupt, and there is no lithologic change across the transition, although the polarity change does occur between samples and not in the middle of a continuous piece. This is not unexpected since the rock is fairly hard and the longest single piece recovered was only 17.5 cm long.

The reversed zone appears to end approximately 1.5 m above the base of the hole where a very short normal occurs, followed by a reversed interval almost 1 m thick and then by a ½-m-thick normal zone at the base of the hole. The two normal samples in the lower reversed zone and one of the reversed samples in the lower normal zone are anomalous single measurements; at other temperatures the samples were reversed and normal. In most normal multiple-core sampling programs the thin lower zones would probably be missed because of the sampling interval or would be considered as single anomalous samples if they were sampled.

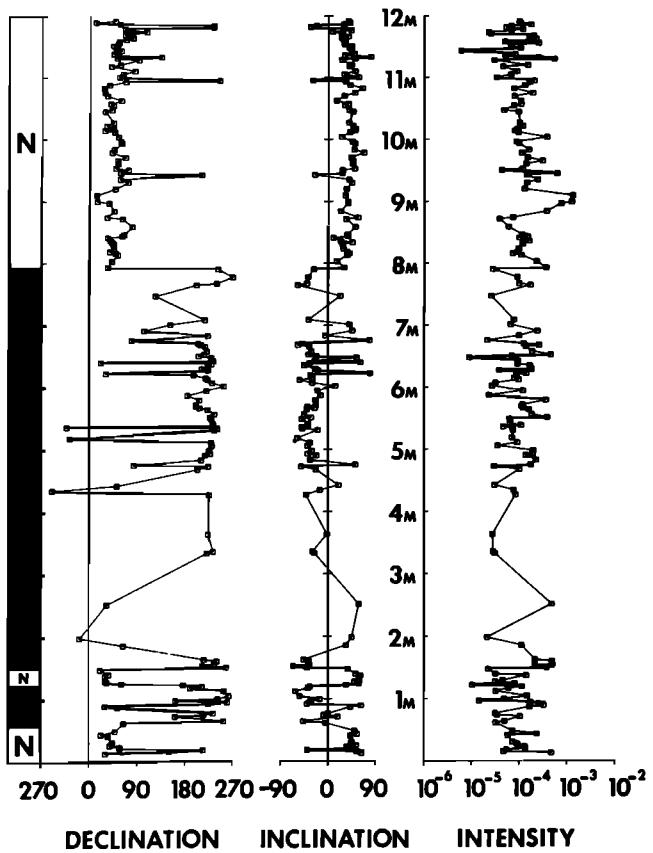


Fig. 6. Structurally corrected stratigraphic plot of declination, inclination, and intensity of remanent magnetization of all samples demagnetized at 627° or 647°C. Intensity plots give total sample intensity in electromagnetic units (12-cm³ samples). A paleomagnetic pole position calculated from all the samples shown in this figure (181) is shown as the solid triangle in Figure 2. It is located at 99.8°E, 34.0°N. Fisher's estimate of precision k = 10.23, and α_{95} = 3.44.

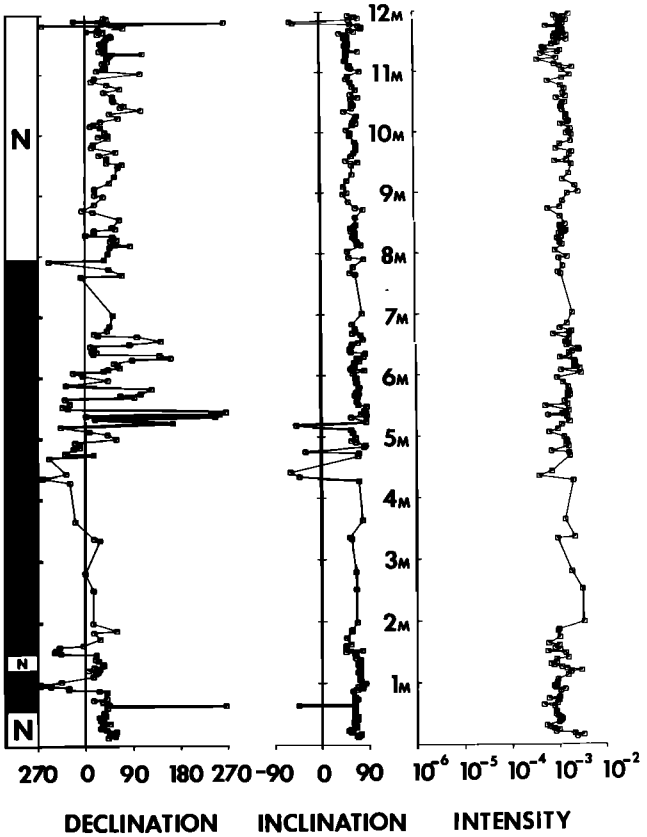


Fig. 7. Structurally corrected stratigraphic plot of declination, inclination, and intensity of NRM directions for all samples studied. Intensity plots give total sample intensity in electromagnetic units (12-cm³ samples).

If the short normals are real and do not span unrecognized hiatuses, they represent some of the shortest magnetic zones yet found. *Steiner and Helsley* [1974] have stated that reversals found in the Kayenta formation in Utah indicate a higher reversal frequency during the uppermost Triassic in comparison with the older parts of the Triassic. If the two normals do exist, the upper short normal and the short reversed zone indicate that reversal frequency may have been high during Carnian time also.

Summary

Results from a 12.2-m-long core collected from the Malmros klint member of the Fleming Fjord (formerly Cape Biot) formation confirm the presence of reversals initially reported by *Bidgood and Harland* [1961]. The reversal sequence contains three zones of normal polarity and two reversed polarity zones.

The presence of a fairly stable secondary component may be due to a Tertiary heating event. In this light the agreement of *Bidgood and Harland*'s [1961] pole for the Cape Biot formation at 72°N, 150°E with *Tarling*'s [1966] east Greenland Tertiary igneous pole at 63°N, 174°E probably indicates that Bidgood and Harland's samples also contain large Tertiary secondary magnetizations.

A pole for the Fleming Fjord formation at 104°E, 34°N is in poor agreement with its expected position based on rotated North American Upper Triassic poles and is only in very rough accord with its expected position based on similar

average Triassic poles from Europe. A colatitude determination is in good agreement with the European colatitude of *Roy* [1972] and suggests that Greenland was much further south of its present position in the Upper Triassic, a position more consistent with the deposition of red beds and evaporites during this period. The data are also consistent with *Roy*'s [1972] hypothesis for right lateral strike slip faulting between Greenland and North America prior to the Upper Triassic.

In conclusion, we regard both Bidgood and Harland's study and our own studies as being of a reconnaissance nature and feel that a thorough study of the Triassic of Jameson Land would answer a host of paleomagnetic questions about the tectonic history of the North Atlantic and provide a reversal sequence that would span the whole of the Triassic.

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